COMPX221 Practical Test 1 2026

Assessment

10	GardenObject class implemented correctly 1 mark for abstract class and 2 marks for fields 3 marks for constructor 3 marks for abstract methods 1 mark for toString() method
10	Flower class implemented correctly 1 mark for inheritance and 1 mark for fields 3 marks for constructor 2 marks for toString() method 2 marks for drawObject() method 1 mark for growObject() method
10	Hedgehog class implemented correctly 1 mark for inheritance and 1 mark for fields 3 marks for constructor 2 marks for toString() method 1 marks for drawObject() method 1 mark for growObject() method 1 mark for moveObject method
10	Loading the csv file of data correctly and adding objects to the list
2	Data for all objects is displayed in the console window
1	Drawing all objects to the sketch window
2	New objects can be added using the mouse buttons
2	Hedgehog objects move around the sketch window
3	Random objects will grow in size
	Total

Getting Started

To begin, download the zip file of the template folder from Moodle and unzip the file. Then add your first name to the end of the folder name. Go into the folder and into the PracticalTest1 folder and then open the sketch in processing. At the end of the test you need to zip up the folder that has your name in it and submit that to the Prac Test 1 Submission activity and make sure you click the submit button.

Test Conditions

Practical Test 1 will be an open book test. You can refer to any printed notes, look at information on Moodle as well as use the internet. You have 105 minutes to complete as much as you can.

You are not allowed to use any of the following during the test:

- email applications or file sharing applications
- any AI tools that generate processing code like chat GPT, etc
- facebook or other social media websites, online chat applications
- Any other applications which allows communication with other people.

If you have managed to get the answers for this test, then you are not allowed to refer to them during the test.

BackGround

You have been asked to help work on a game for small children to help them learn about the importance of gardening. The player will be able to add flowers to the garden and also hedgehogs. The hedgehogs will move around randomly and eat any flowers they come in contact with. Nilesh has asked you to model the flowers and hedgehogs as he thinks about the mechanics of the game. Take a look at the class diagram provided inside the template sketch folder.

Some code in the sketch tab is provided for you test your classes before processing the CSV file of data. You should just comment and uncomment the code as required when testing.

A csv file has been provided for you called data.csv. It has 2 different types of lines in the file.

A hedgehog line has the following format (all values are integers):

```
x,y,width,height
```

A flower line has the following format (all values are integers):

```
x,y,size
```

Tasks

1. Implement the GardenObject class (**which is an abstract class**) as specified in the class diagram using the notes below:

   The constructor will create a new PVector object for the position using the x and y values passed in. The PVector object is the centre of the object.

   The drawObject(), moveObject() and growObject() methods are all abstract methods.

   The toString() method returns back the x and y position as a string with a comma between both values (i.e. 30,50).

2. Implement the Flower class as specified in the class diagram using the notes below:

   The constructor will set the position to the x and y values passed in and also set the object's colour to `#57CE13`.

   The toString() method will return the word "Flower:" as well as the x and y position and the size of the flower to the console window in the following format:

   ```
   Flower: x,y size
   ```

Note: Think carefully how to write the code so that there is no repeated code.

The drawObject() method should draw a circle for the flower centred around it's x and y position, in the correct size and colour.

The moveObject() method should be created but leave it empty as Nilesh doesn't want the flowers to move.

The growObject() method should make the size of the flower increase by 5 pixels.

Test your class by creating and displaying some flowers by uncommenting the provided code in the sketch tab. Once they are displaying correctly comment out the code again.

3. Implement the Hedgehog class as specified in the class diagram using the notes below:

The constructor will set the position to the x and y values passed in and also set the colour to `#6C510B`. Then also set the width and height of the hedgehog to the values passed in.

The toString() method will return the word "Hedgehog:" as well as the x and y position and the width and height of the hedgehog to the console window in the following format:

```
Hedgehog: x,y width height
```

Note: Think carefully how to write the code so that there is no repeated code.

The drawObject() method should draw a rectangle for the hedgehog centred around it's x and y position, in the correct width and height and colour.

The moveObject() method should move the hedgehog by the amount passed to the method in a random direction.

Hint: Generate a random number between 1 and 4 as an integer. If the number is a 1 move up, if the number is a 2 move right, if the number is a 3 move down and if the number if a 4 move left.

Hint: int num = (int)random(1,8) will generate a random integer between 1 and 7 inclusive.

The growObject() method should make the width and height of the hedgehog increase by the amount passed to the method.

Test your class by creating and displaying some Hedgehogs by uncommenting the provided code in the sketch tab. Once they are displaying correctly comment out the code again.

4. Create an array list at class scope level named objectList in the sketch tab. Think carefully what the datatype of the array list should be so that you can store objects from both subclasses in the list.

5. In the setup method write the code to load all the data from the csv file and create the appropriate objects from the sub classes and store all of the objects in the objectList.

6. After loading the data from the csv file display information from each object to the console window.

Note: Don't worry if the x and y values are displayed float values.

7. In the draw method in the sketch tab, write the code that will draw all of the objects to the sketch window. Test that all of the objects are displayed. Then modify the code so that after drawing the object it is moved. Test that your hedgehog objects now move (the flowers won't move).

8. In the sketch tab, go to the mousePressed() method and write the code so that if the left button is pressed a flower is created at the mouse position with a size of 30 and is added to the object list.

   If the right mouse button is pressed then write the code to create a new hedgehog at the mouse position with a width of 20 and a height of 10 and is added to the object list. Test that you can now create new objects when pressing the mouse buttons.

9. At the bottom of the draw() method in the sketch tab, write the code to randomly make an object from the object list grow.

Hint: Make sure the list is not empty before you try and make an object grow.